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STANDARD OPERATING PROCEDURE: STRUCTURAL STEEL CAPACITY & SAFETY

Document ID: SOP-ENG-2026-001 | Revision: 3.2 | Effective Date: Jan 2026

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1. PURPOSE & SCOPE

This standard operating procedure mandates safety parameters and mathematical verification models for structural steel members deployed in high-load industrial infrastructures. Compliance with IS-800 and ASTM A36 standards is mandatory across all design and execution phases.

2. MATERIAL CONSTANTS (ASTM A36 STEEL)

- Yield Strength (F_y): 250 MPa (N/mm²)
- Elastic Modulus (E): 200,000 MPa
- Density: 7,850 kg/m³
- Poisson's Ratio: 0.26

3. SAFETY & ALLOWABLE STRESS LIMITS

Under standard structural load conditions, allowable bending stress must not exceed 66% of the material yield strength:

- Formula: $\text{Allowable_Stress} = 0.66 * F_y$
- Baseline Limit for Grade A36: $\text{Allowable_Stress} = 165.0 \text{ MPa}$

4. MATHEMATICAL FORMULATION FOR SIMPLY SUPPORTED BEAMS

To evaluate beam capacity under a central point load:

1. Calculate Applied Force (F): $\text{Force_N} = \text{Mass_kg} * 9.81$
2. Convert Beam Span (L) to millimeters: $\text{Span_mm} = \text{Span_m} * 1000$
3. Calculate Maximum Bending Moment (M): $\text{Moment_Nmm} = (\text{Force_N} * \text{Span_mm}) / 4$
4. Calculate Actual Bending Stress (σ): $\text{Bending_Stress_MPa} = \text{Moment_Nmm} / \text{Section_Modulus_Z_mm}^3$

5. VERIFICATION & EVALUATION RULE

- IF $\text{Bending_Stress_MPa} \leq 165.0 \text{ MPa}$: PASS (Structure is safe for operation).
- IF $\text{Bending_Stress_MPa} > 165.0 \text{ MPa}$: FAIL (Exceeds safety capacity; redesign required).